

Predictive Analytics for Risk Profiling and Triaging COVID-19 Patients

Final Report_G2

Armando Vizcardo-Benites

Daria Nefedova

Kretee Arora

Maria Navarro

Shiji Valsan

Background

The Smart ICU Bed Allocation System (SIBAS) is a predictive analytics tool designed to address the critical need for efficient allocation of intensive care unit (ICU) beds for COVID-19 patients.

We have utilized various models and data analytic techniques to develop the most optimal predictive risk model.

Methodology

The dataset comprises 370,633 patients diagnosed with COVID-19 (ICD-10 code U071), which serves as the anchor diagnosis but it's not a feature for the predictive model.

Our focus remains on DX1 to DX20, along with other demographic variables like age and gender, to predict mortality. The dataset is populated with 227 variables based on ICD-10 codes, coded as binary variables.

1. Files Used

- a) Covid19_SourceFile.XLSX: Patient data diagnosed with COVID-19
- b) ICD10_Dx_TaxonomyTree.XLSX: 69,000 diagnosis

2. Analytical Approach Used

- a) Data Preparation
- b) Model Development
- c) Validation & Iteration
- d) Addressing the class Imbalance
- e) Documentation & Reporting

3. Analytical Approach Used

- a) Using entries in DGL_3_Extend, we have reduce the number of diagnosis from 69,000 to 227 binary variables
- b) We then have split the data into train (75%) and test (25%) for all PCA & models
- c) Performed a PCA analysis to understand the data
- d) Setting up hyperparameters or configuration details for each model

Model	Hyperparameters
Logistic Regression	penalty: None, l2, l1 C: 0.001, 0.01, 0.1, 1, 10, 100 solver: saga, lbfgs
Random Forest	n_estimators: 100
Random Forest 300	n_estimators: 300
XGB Classifier	n_estimators: 100, 500, 1000 max_depth: 3, 5, 7
Gradient Boosting Classifier	n_estimators: 100

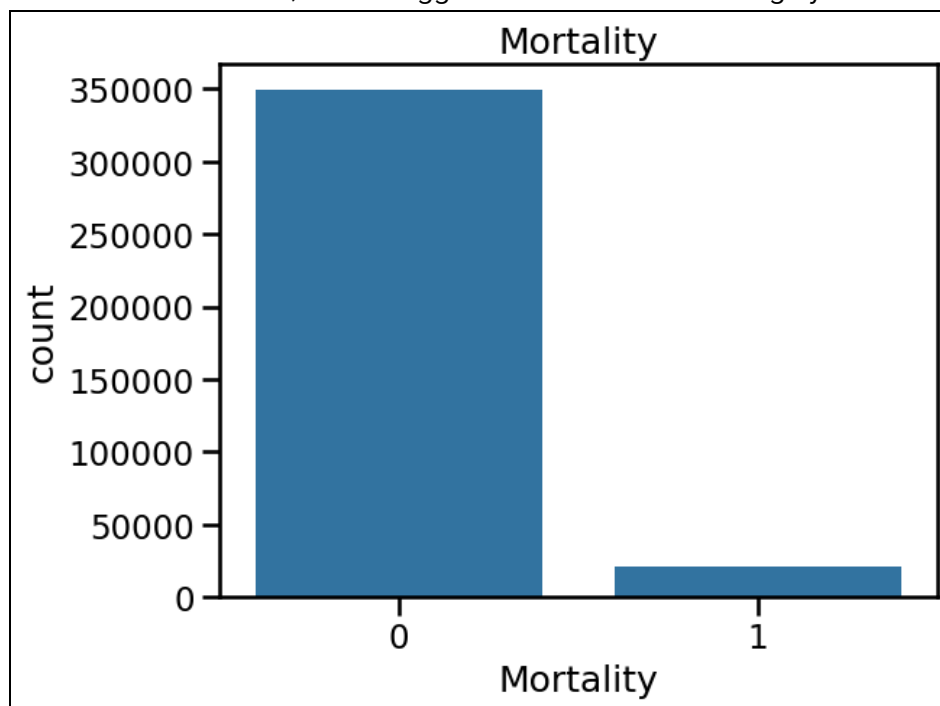
- e) Conducted grid search with cross validation to find the best combination of hyperparameters for a given model and ensuring that each fold has the same class distribution as the original dataset. This gives us a better idea of the performance of unseen data, generalizability of each respective model.
- f) Testing for 4 ML models: logistic regression, random forest, XGB classifier, and gradient boosting classifier.
- g) Using precision, recall and F1 scores to evaluate the best model. We identified the best performing model based on the specified evaluation metric (F1 score). This is done by selecting the model with the highest F1 score across all combinations of hyperparameters. When dealing with imbalanced datasets, traditional accuracy metrics can be misleading because they favor the majority class, leading to models that perform poorly on the minority class. In such cases, the F1 score is a better metric for evaluating model performance. The F1 score is the harmonic mean of precision and recall. Precision measures the proportion of true positive predictions among all positive predictions, while recall measures the proportion of true positive predictions among all actual positive instances. By considering both precision and recall, the F1 score provides a balanced assessment of a model's performance on both the positive and negative classes. Since the F1 score considers both false positives and false negatives, it is well-suited for imbalanced datasets where the classes have unequal distributions. It penalizes models that make too many false positives or false negatives, making it a more reliable metric for evaluating model performance in such scenarios. The F1 score combines precision and recall into a single metric, making it easy to compare the performance of different models or parameter settings.
- h) Plotting graphs for ROC and Precision

- i) To address class imbalance we have used random under sampling, reduced and created a dataset with the number of mortality=0 ('survived') equal to mortality=1 ('Died'). We created similar models on this dataset and evaluated the results as for the original dataset

Understanding the Data

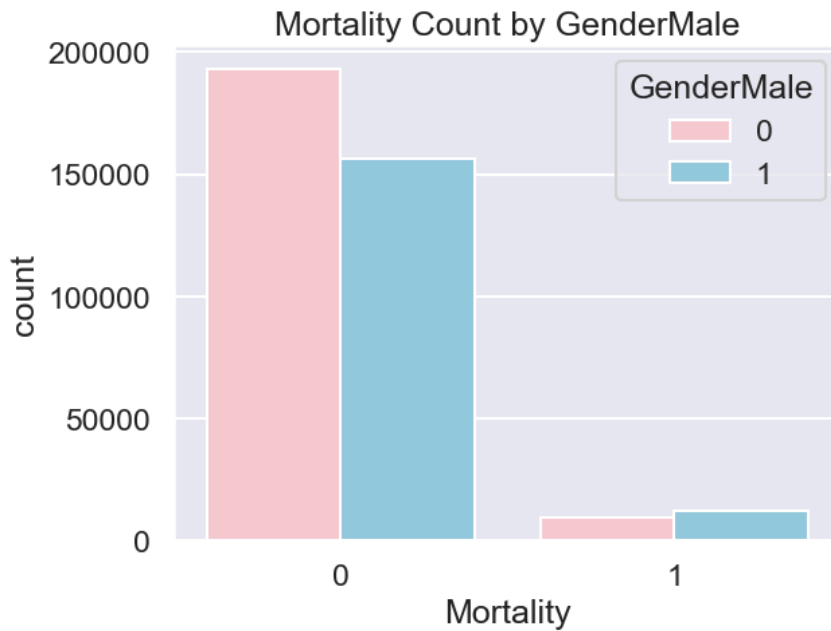
1. Mortality

- a) Binary var
- b) The plot below shows the count of mortality class in our dataset
- c) 94% survived and 6% died, which suggests that the dataset is highly imbalanced



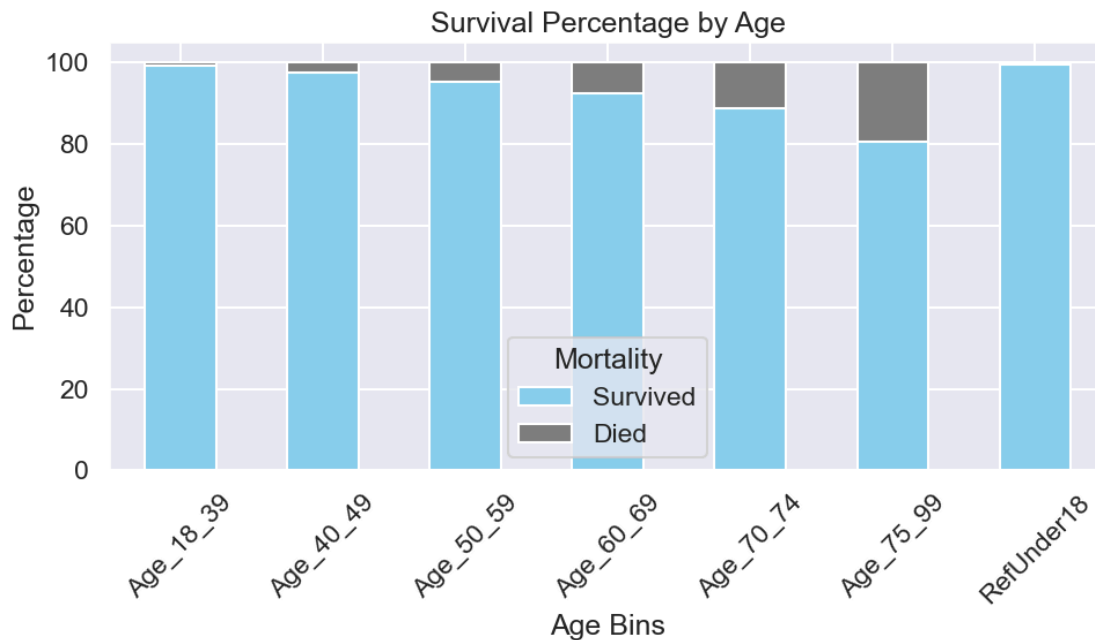
2. Mortality by Gender

- a) The plot below shows the count of mortality count by gender in the dataset
- b) Women are more likely to survive the COVID-19 infection than men



3. Mortality & Survival by Age

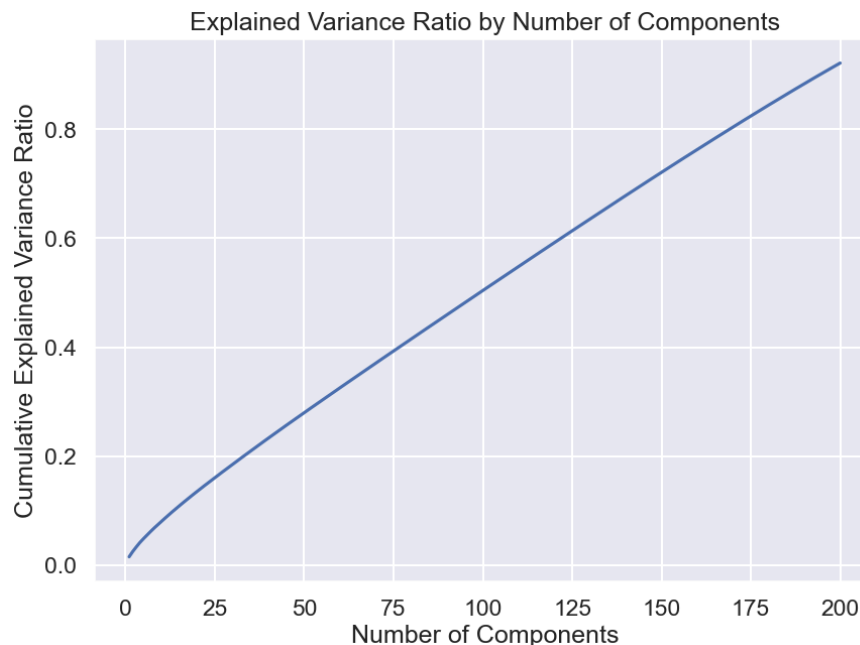
- The plot below shows the percentage of mortality & survival by age category in the dataset
- As the age increases, the percentage of patients who died increases (refunder18 column is used as a baseline)



Analysis & Models

Principal Component Analysis (PCA)

- a) Performed to reduce the dimensionality of the dataset
- b) Dataset was split into train (75%) and train (25%)
- c) Scaling of features to ensure each feature contribute equally to the variance
- d) The following plot shows the explained variance ratio by number of components using PCA. The plot of explained variance ratio by the number of components appears to be linear. This suggests that each additional principal component contributes roughly the same amount of variance to the total variance explained:
 1. Linear Increase in Explained Variance: Each principal component captures a similar amount of variance in the data, resulting in a linear increase in the cumulative explained variance ratio as the number of components increases. This suggests that each additional component adds valuable information to the representation of the data.
 2. No Clear Dominant Components: In some cases, there may not be dominant components that explain a significantly larger proportion of variance compared to others. Instead, the variance is distributed relatively evenly across multiple components, leading to a linear trend in the explained variance ratio plot.



Logistic Regression

This is a statistical method used for analyzing a dataset in which there are one or more independent variables that determine an outcome. The outcome is measured with a variable in which there are only two possible outcomes.

Advantages:

1. Interpretable: Easy to understand the relationship between input variables and the probability of the outcome.
2. Fast training and prediction: Computationally efficient for large datasets.
3. Well-suited for binary classification tasks.

Disadvantages:

1. Assumes linear relationship: May not capture complex nonlinear patterns in the data.
2. Limited expressive power: Not as flexible as other models like decision trees or neural networks.

Random Forest

This model is made up of many decision trees which uses two major concepts: (i) random sampling of training data points when building the trees and, (ii) random subsets of features considered when splitting the nodes. “Gini” criterion is used to build the model.

Advantages:

1. Excellent performance: Often provides high accuracy and generalization.
2. Handles large datasets with high dimensionality well.
3. Robust to overfitting: Aggregates multiple decision trees, reducing the risk of overfitting.

Disadvantages:

1. Less interpretable: Not as easy to interpret as simple models like logistic regression.
2. Slower training: Can be computationally expensive for very large datasets.

XGBoost Classifier

XGBoost is a decision-tree-based ensemble machine learning algorithm that uses a gradient boosting framework. In prediction problems involving unstructured data (e.g., images, text, etc.) artificial neural networks, it tends to outperform all other algorithms or frameworks. It is an implementation of gradient boosted decision trees designed for speed and performance. The



goal of this library is to push the limits of the computation limits of machines to provide scalable, portable and accurate data.

Advantages:

1. State-of-the-art performance: Often achieves high accuracy and generalization.
2. Scalable: Optimized for speed and efficiency, can handle large datasets.
3. Built-in regularization: Helps prevent overfitting.

Disadvantages:

1. Requires tuning: Hyperparameters need to be carefully tuned for optimal performance.
2. Less interpretable: Can be difficult to interpret the inner workings of the model.

Gradient Boosting Classifier

Gradient boosting is a machine learning technique for regression and classification problems, which produces a prediction model in the form of an ensemble of weak prediction models, typically decision trees.

Advantages:

1. Excellent performance: Similar to XGBoost, gradient boosting often achieves high accuracy.
2. Robust to overfitting: Uses boosting to iteratively improve performance without overfitting.
3. Handles different types of data well.

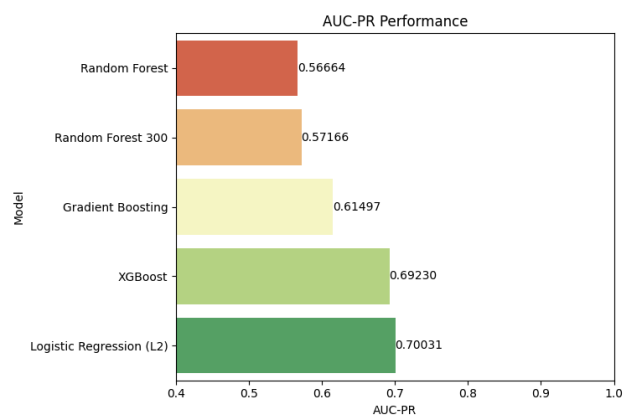
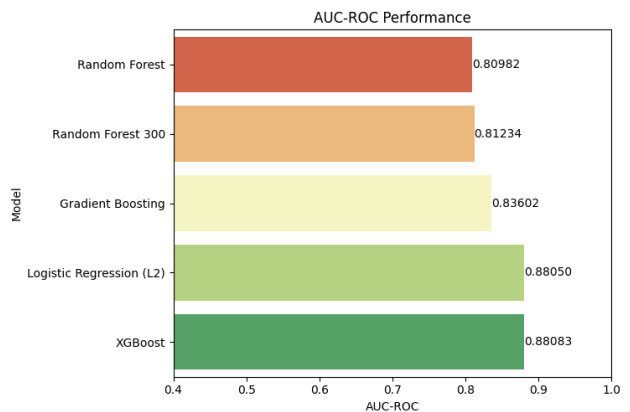
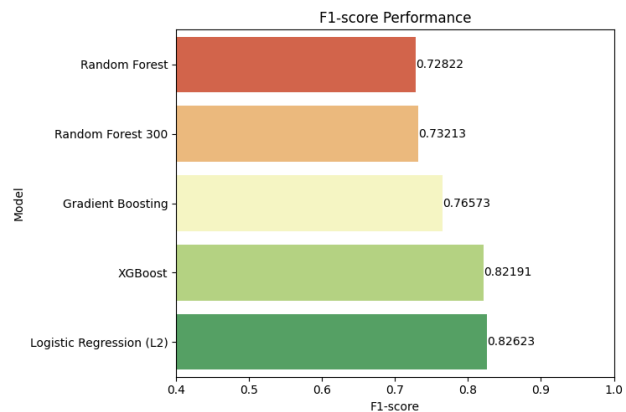
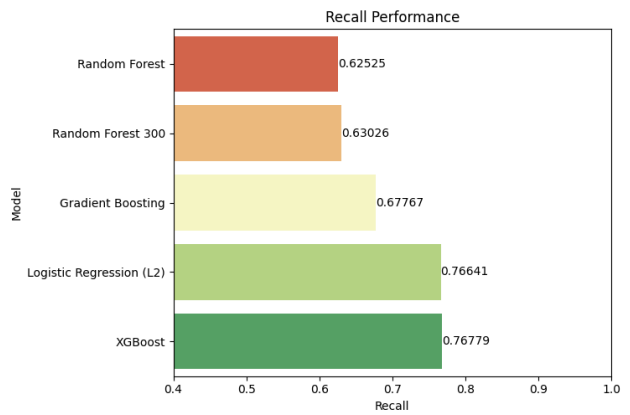
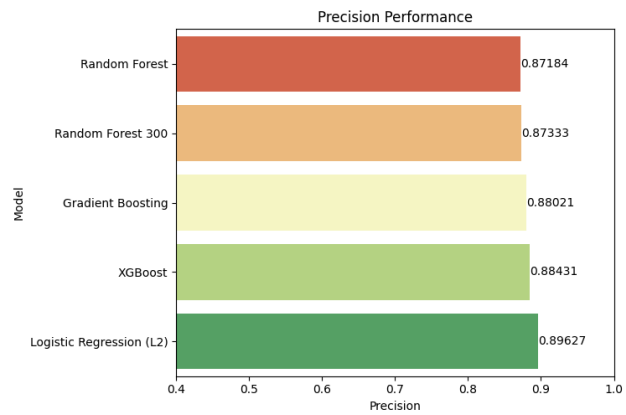
Disadvantages:

1. Slower training: Like random forests, gradient boosting can be computationally expensive.
2. Requires tuning: Hyperparameters need to be optimized for best results.
3. Less interpretable: Similar to XGBoost, the inner workings of the model may be difficult to interpret.

Results

We calculated precision, recall, f1 score, area under the ROC curve and the area under the precision and recall curve.

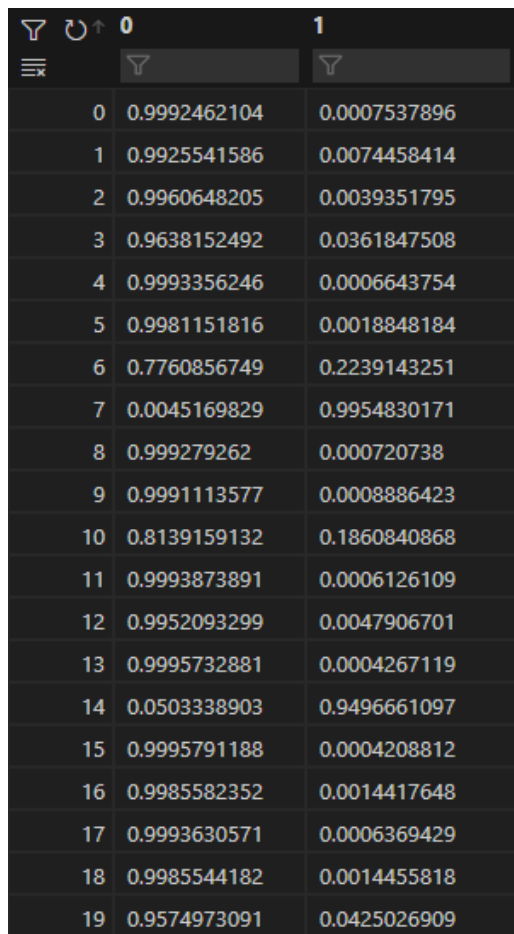
	Precision	Recall	F1-score	AUC-ROC	AUC-PR
Model					
Logistic Regression (L2)	0.896266	0.766408	0.826231	0.880500	0.700308
Random Forest	0.871840	0.625251	0.728219	0.809824	0.566642
Random Forest 300	0.873332	0.630261	0.732133	0.812344	0.571659
XGBoost	0.884309	0.767786	0.821915	0.880830	0.692303
Gradient Boosting	0.880210	0.677668	0.765726	0.836021	0.614973



Based on our analysis, **our best model is Logistic Regression.**

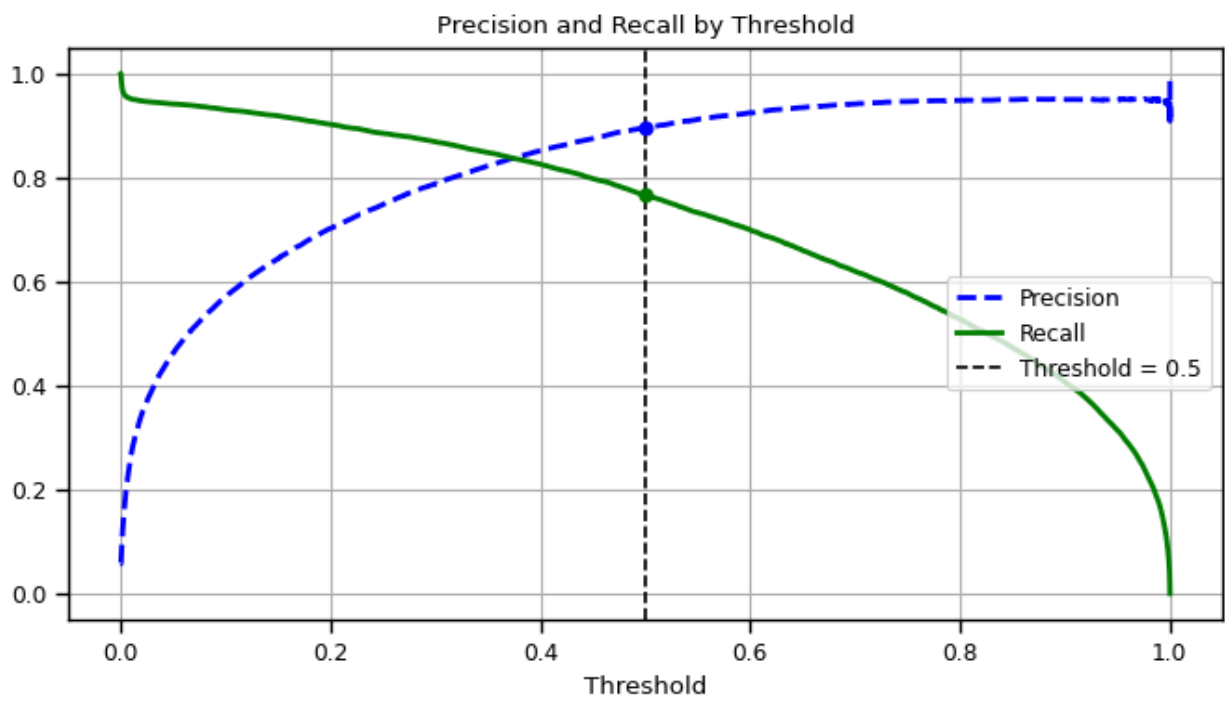
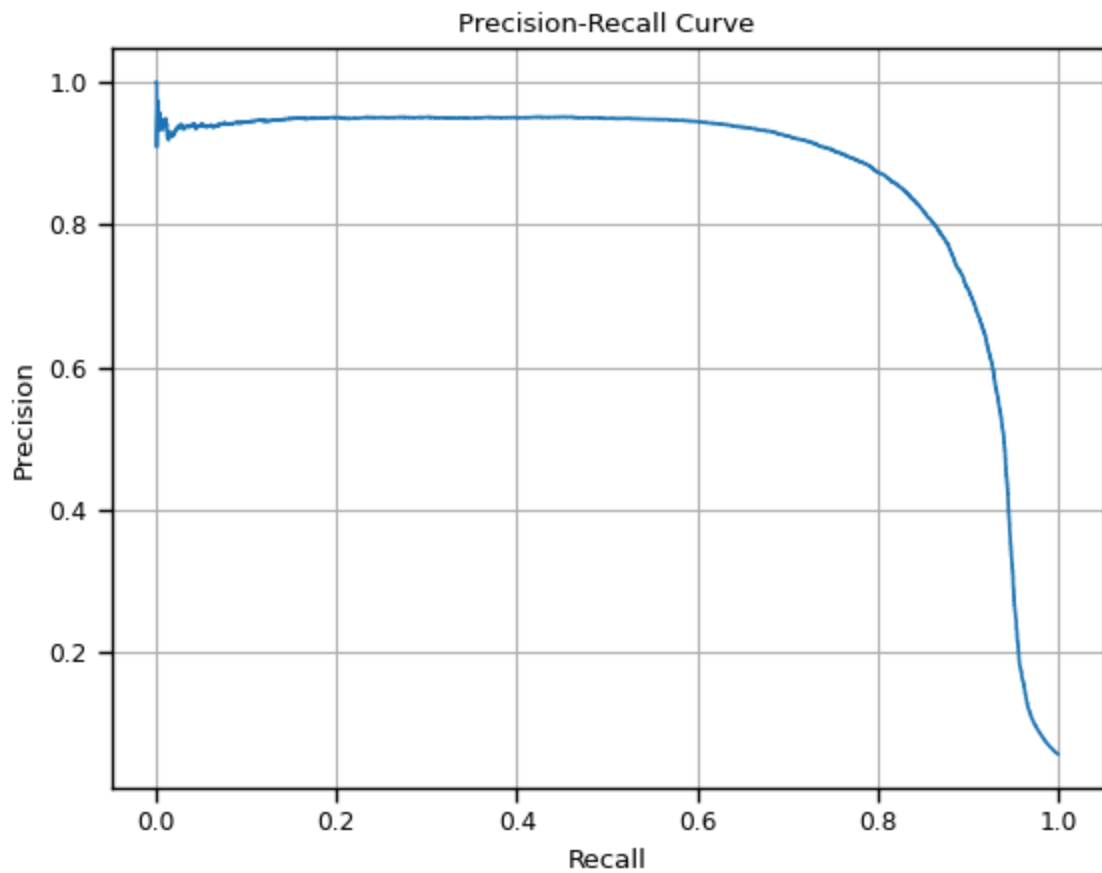
Although the two best performing models are Logistic Regression (with Ridge regularization) and XGBoost Classifier. Logistic Regression has the highest F1 score, precision and AUC ROC, so we chose it as our best predictive model.

Instead of using the binary predictions directly, we can obtain the predicted probabilities from the best model. These represent the likelihood of belonging to that class. Once we have the predicted probabilities for the positive class (mortality = 1), we can convert them into risk scores. We could use the predicted probabilities as the risk scores directly. For example, a predicted probability of 0.8 for mortality might correspond to a risk score of 80%. Here is a screenshot of some of the predicted probabilities for the training set of the best model:



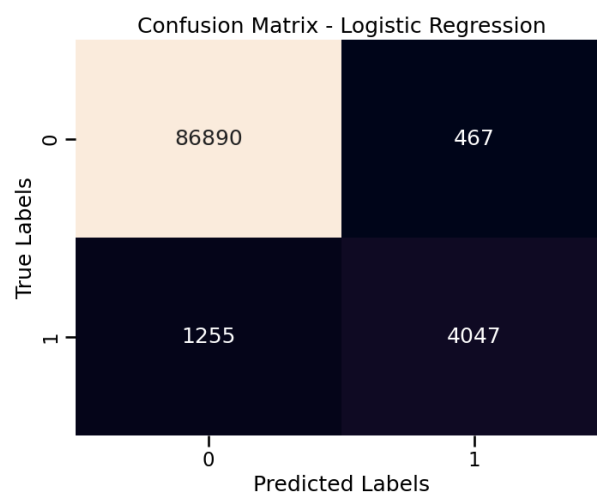
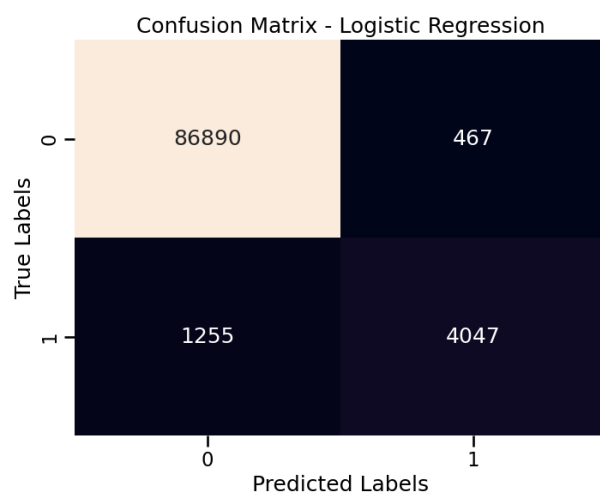
	0	1
0	0.9992462104	0.0007537896
1	0.9925541586	0.0074458414
2	0.9960648205	0.0039351795
3	0.9638152492	0.0361847508
4	0.9993356246	0.0006643754
5	0.9981151816	0.0018848184
6	0.7760856749	0.2239143251
7	0.0045169829	0.9954830171
8	0.999279262	0.000720738
9	0.9991113577	0.0008886423
10	0.8139159132	0.1860840868
11	0.9993873891	0.0006126109
12	0.9952093299	0.0047906701
13	0.9995732881	0.0004267119
14	0.0503338903	0.9496661097
15	0.9995791188	0.0004208812
16	0.9985582352	0.0014417648
17	0.9993630571	0.0006369429
18	0.9985544182	0.0014455818
19	0.9574973091	0.0425026909

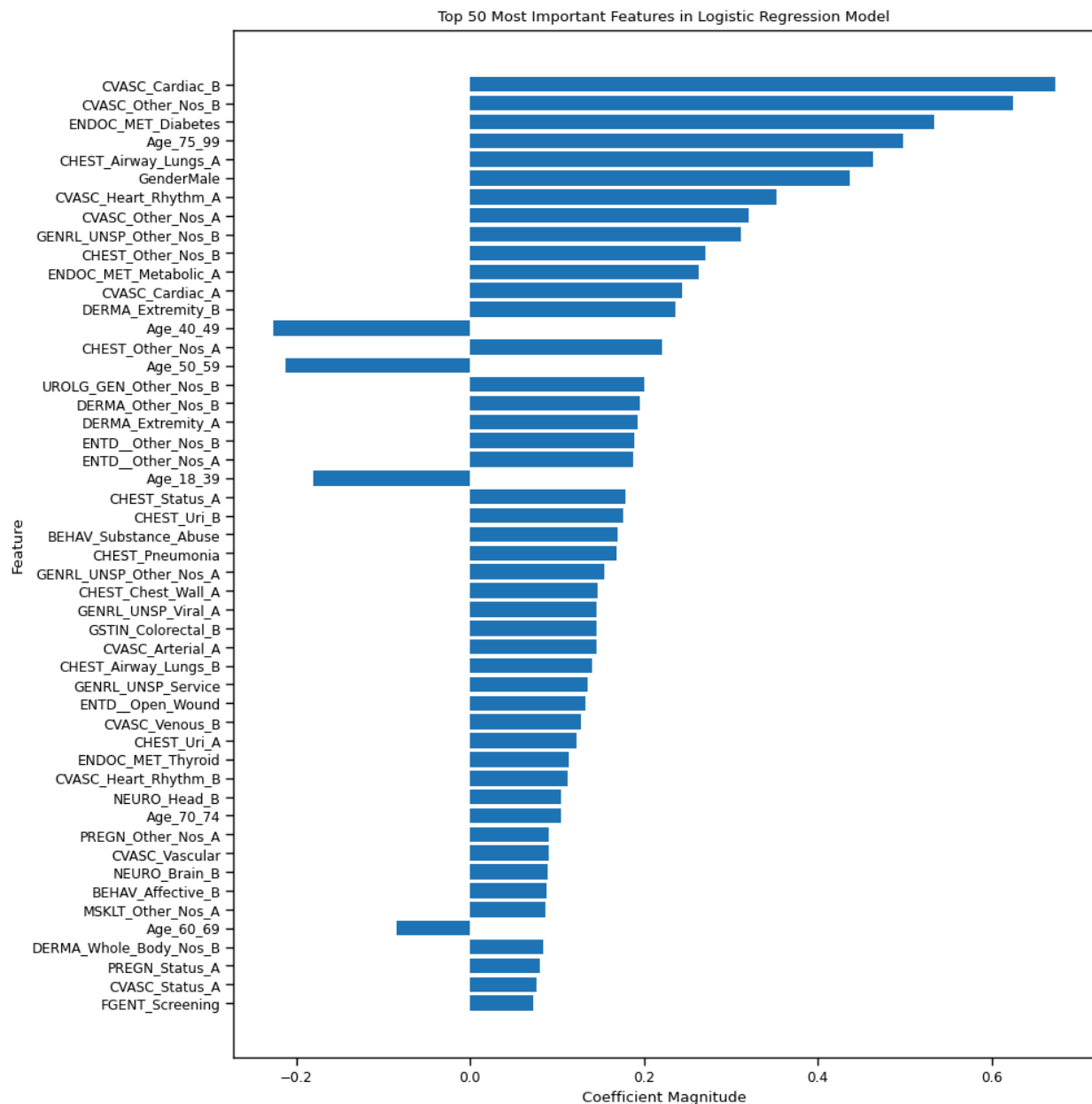
We have also illustrated how precision and recall change depending on the threshold we chosen. The threshold is used to convert the continuous risk scores into binary predictions. This threshold can be chosen based on the desired balance between precision and recall.



We took the best performing model (based on F1 score) to make predictions on the test set and created confusion matrices for the two best performing models:

	Precision	Recall	F1-score	AUC-ROC	AUC-PR
Model					
Logistic Regression (L2)	0.899556	0.763485	0.825954	0.879156	0.700331
Random Forest	0.877404	0.645228	0.743615	0.819878	0.586426
Random Forest 300	0.872276	0.649189	0.744377	0.821710	0.586345
XGBoost	0.886562	0.766503	0.822173	0.880275	0.692913
Gradient Boosting	0.883767	0.679743	0.768443	0.837159	0.619060





This plot will show you the relative importance of each feature in the logistic regression model. Features with larger absolute coefficient values are considered more important in predicting the target variable. Since we can analyze the importance of features in a logistic regression model by examining the coefficients associated with each feature. As logistic regression is a linear model, the magnitude of the coefficients provides information about the importance of each feature in predicting the target variable, which can be seen on the given plot.

Based on the findings of our logit model, we have created word clouds with all the DGLs and then by further reducing the diagnosis level used in the regression to major diagnostic categories (MDCs) along with the demographics.

Addressing Class Imbalance

As mentioned earlier, we have tried to address the issue of class imbalance in the dataset by randomly undersampling the mortality=0 (survived). The final class distribution looks like follows:

Mortality	Training Dataset	Test Dataset
0	15952	87341
1	15952	5318

We ran the same four models with different and most optimal hyperparameters. Since this dataset was smaller, we were able to run more model configurations than in the previous dataset with many more observations.

Model	Hyperparameter
Logistic Regression (L2 regularization)	'penalty': ['l2'] 'C': [0.001, 0.01, 0.1, 1, 10, 100], 'solver': ['lbfgs', 'sag', 'saga']
Logistic Regression (L1 regularization)	'penalty': ['l1'] 'C': [0.001, 0.01, 0.1, 1, 10, 100] 'solver': ['liblinear']
Random Forest	'n_estimators': [100]
Random Forest 300	'n_estimators': [300]
Random Forest 500	'n_estimators': [500]
Random Forest 1000	'n_estimators': [1000]
XGB Classifier	'n_estimators': [100, 500, 1000] 'max_depth': [3, 5, 7]
Gradient Boosting Classifier	'n_estimators': [100]

We obtained the following results for our training dataset:

	Precision	Recall	F1-score	AUC-ROC	AUC-PR
Model					
Logistic Regression (L2)	0.942544	0.936685	0.939598	0.939788	0.914522
Logistic Regression (L1)	0.942812	0.936121	0.939447	0.939663	0.914521
Random Forest	0.936015	0.917502	0.926648	0.927376	0.900033
Random Forest 300	0.937004	0.918693	0.927737	0.928442	0.901462
Random Forest 500	0.936609	0.919195	0.927799	0.928473	0.901316
Random Forest 1000	0.936915	0.919257	0.927983	0.928661	0.901624
XGBoost	0.946127	0.932422	0.939217	0.939663	0.915980
Gradient Boosting	0.929641	0.925714	0.927656	0.927815	0.897715

This shows Logistic Regression (with L2 regularization) and XGBoost perform the best. The scores for test dataset:

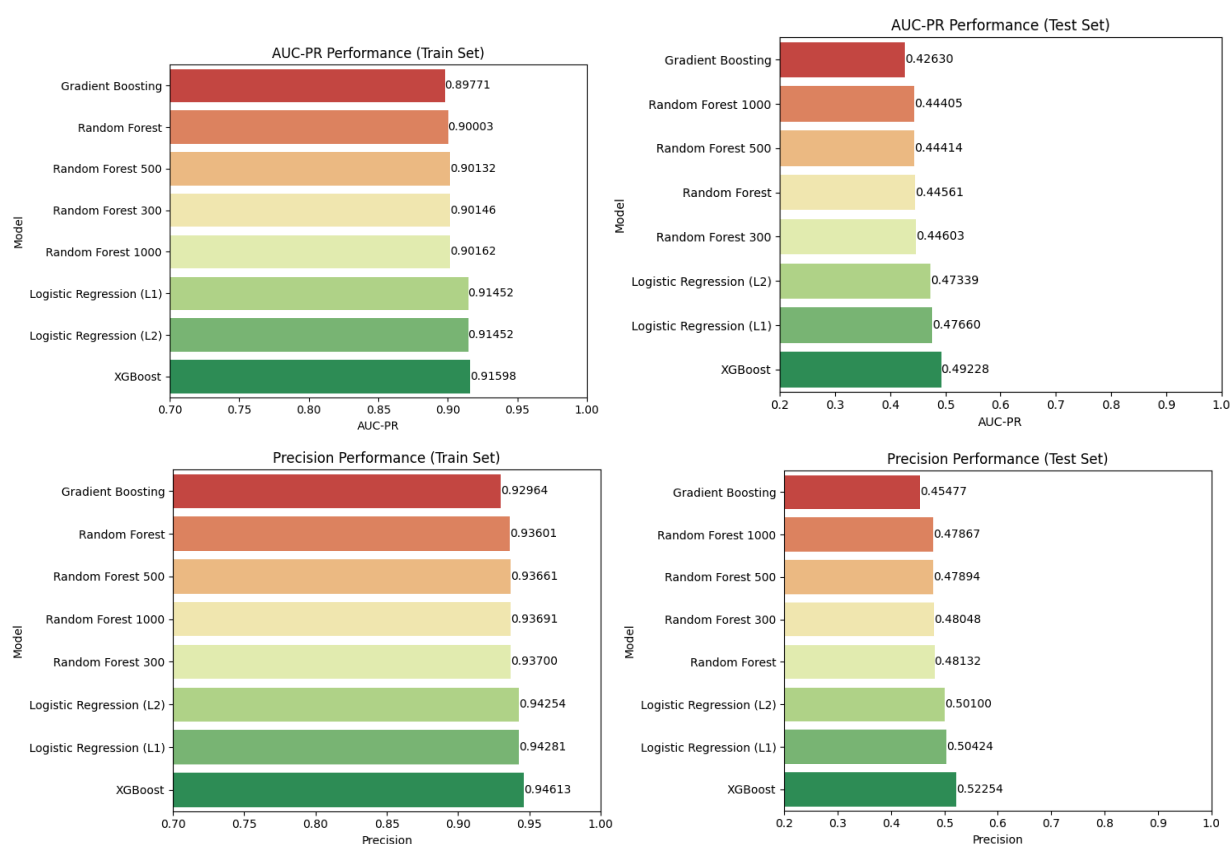
	Precision	Recall	F1-score	AUC-ROC	AUC-PR
Model					
Logistic Regression (L2)	0.501005	0.937759	0.653091	0.940445	0.473394
Logistic Regression (L1)	0.504245	0.938135	0.655930	0.940988	0.476600
Random Forest	0.481320	0.915758	0.630992	0.927836	0.445608
Random Forest 300	0.480476	0.918578	0.630933	0.929051	0.446028
Random Forest 500	0.478944	0.917450	0.629345	0.928338	0.444145
Random Forest 1000	0.478670	0.917826	0.629198	0.928481	0.444052
XGBoost	0.522543	0.934938	0.670397	0.941462	0.492280
Gradient Boosting	0.454772	0.928357	0.610486	0.930294	0.426302

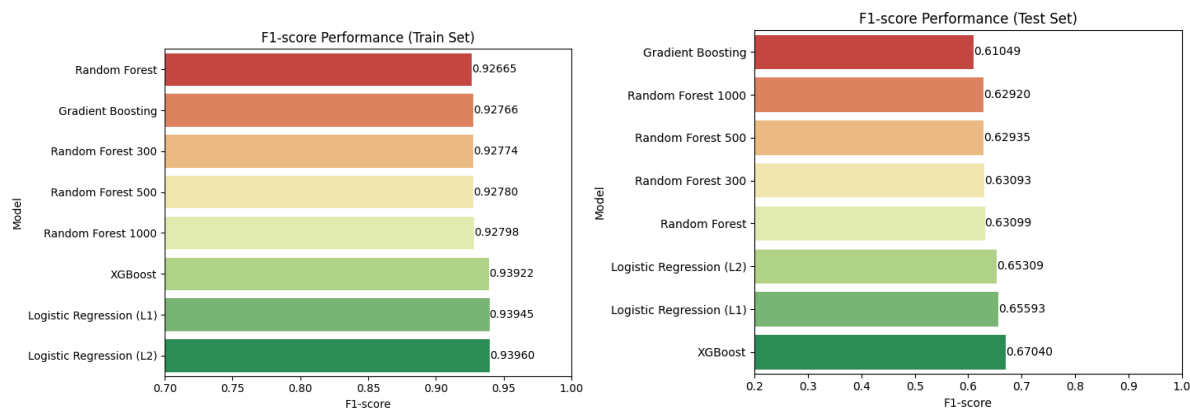
By looking at the results it seems that the models tend to overfit the training data and struggle to generalize on the unseen test data. Precision, F1-score and AUC-PR are significantly lower on the test data.

In a binary classification problem, where one class is considered positive and the other negative, the precision-recall curve illustrates the trade-off between precision (the ability of the classifier not to label a negative sample as positive) and recall (the ability of the classifier to find all positive samples).

However, when comparing our results between balanced and imbalanced datasets, we observe higher scores across all models for the training dataset with balanced data. Conversely, for the test dataset, we notice a lower precision score across all models, while recall is higher. This suggests that with a balanced dataset, we achieve greater sensitivity, which is particularly beneficial in healthcare applications.

An AUC-PR score below 0.50 suggests that the model's precision is lower than what would be achieved by random guessing, indicating poor performance. This could mean that the model is not effectively distinguishing between the positive and negative classes, or it may be misclassifying most of the positive samples. We have compared the AUC-PR, precision and F1 Scores in the train and test sets to highlight the big differences.





Conclusion

The Logistic Regression model, along with XGBoost, showed the highest performance based on metrics such as precision, recall, and F1 score. These metrics indicate the ability of the model to correctly identify positive cases (mortality) and generalize well on unseen data. Although the best performing models for the undersampled dataset were the same, the high performance on the training data, the models exhibited signs of overfitting. This was evident from the significant drop in performance metrics when tested on the unseen test data.

The models also struggled to generalize well on unseen test data. The drop in performance on the test data indicates that the models may not be capturing the true underlying patterns in the data. To improve the generalization of the models, one might use techniques such as cross-validation, hyperparameter tuning, and feature selection to optimize the models. Additionally, more diverse and representative data could help improve the models' ability to generalize to different populations.

The best performing models for the balanced dataset were the same. Comparing results between balanced and imbalanced datasets, we find higher scores for all models in the training dataset with balanced data. In contrast, the test dataset shows lower precision but higher recall with balanced data. This suggests that balanced datasets improve sensitivity, especially valuable in healthcare. We also noted some diagnoses and demographics are more likely to cause mortality with the chief complaint of COVID-19.

However, it is very important to select the model which not only predicts the risk and profiles patients but also mitigates the Type I and Type II error. The risk is not just missing the true positives but also incorrectly predicting the false positives. In the time of crisis, predicting outcome and optimizing resource allocation is critical.

Policy Recommendation

Such predictive models can prove to be a boon in healthcare settings:

1. **Enhancing Triage Protocols:** Integrating the predictive model into triage protocols enables healthcare providers to better identify and prioritize COVID-19 patients at higher mortality



risk, ensuring timely and appropriate care and addressing ethical concerns related to resource allocation.

2. **Clinical Decision Support:** The predictive model serves as a tool for healthcare providers in treatment decisions, improving clinical outcomes and patient care. However, it raises ethical considerations regarding patient autonomy and informed consent.
3. **Informing Public Health Strategies:** Insights from the model inform public health interventions, aiming to reduce COVID-19 mortality rates. This includes targeted interventions for high-risk populations, balancing public health measures with individual rights and privacy.
4. **Facilitating Healthcare Planning & Resource Allocation:** The model assists in planning healthcare needs, estimating demand for resources based on projected cases and mortality rates. This ensures preparedness for case surges and addresses ethical concerns related to equitable access to healthcare.
5. **Promoting Data Sharing and Collaboration:** Encouraging collaboration among healthcare stakeholders improves the accuracy of predictive models. This enhances patient outcomes and policy decisions but requires addressing ethical considerations surrounding data privacy and security.



References

1. Sharma, S., Izzat Alsmadi, Alkhawaldeh, R. S., & Bilal Al-Ahmad. (2022). Data-driven analysis and predictive modeling on COVID-19. *Concurrency and Computation*, 34(28). <https://doi.org/10.1002/cpe.7390>